


# STATS ANALYSIS

An NBA exploratory career statistics analysis

Avg. Reading time: 5 minutes

# DATASET

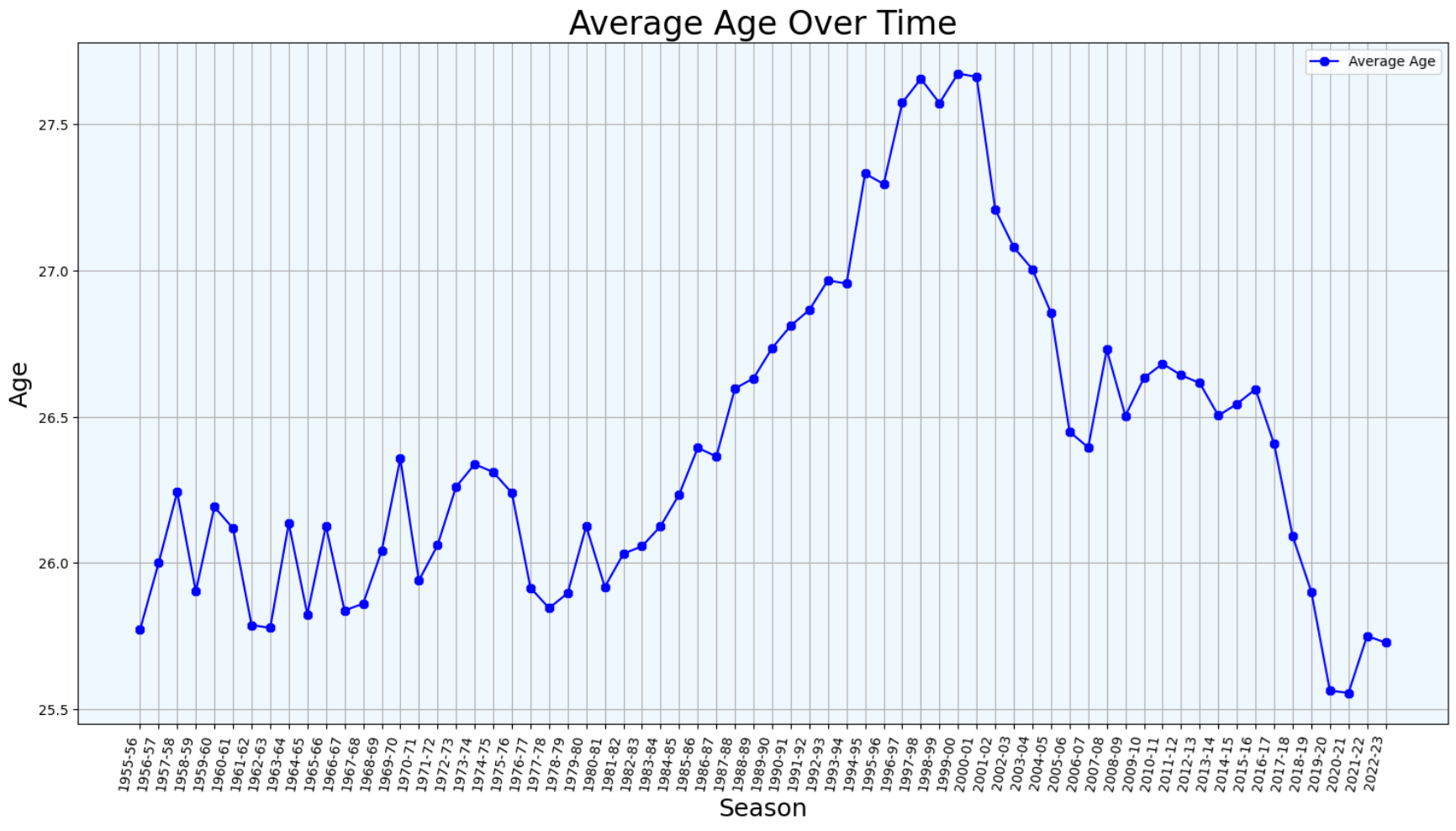
- Data from basketball-reference website
- Use of players per-game and season-total data for every season from 1955-56 to 2022-23
- Total of 27,845 records
- Use of MVP & Championship data
- Implemented in Python on a Jupyter notebook

# DATA PREVIEW

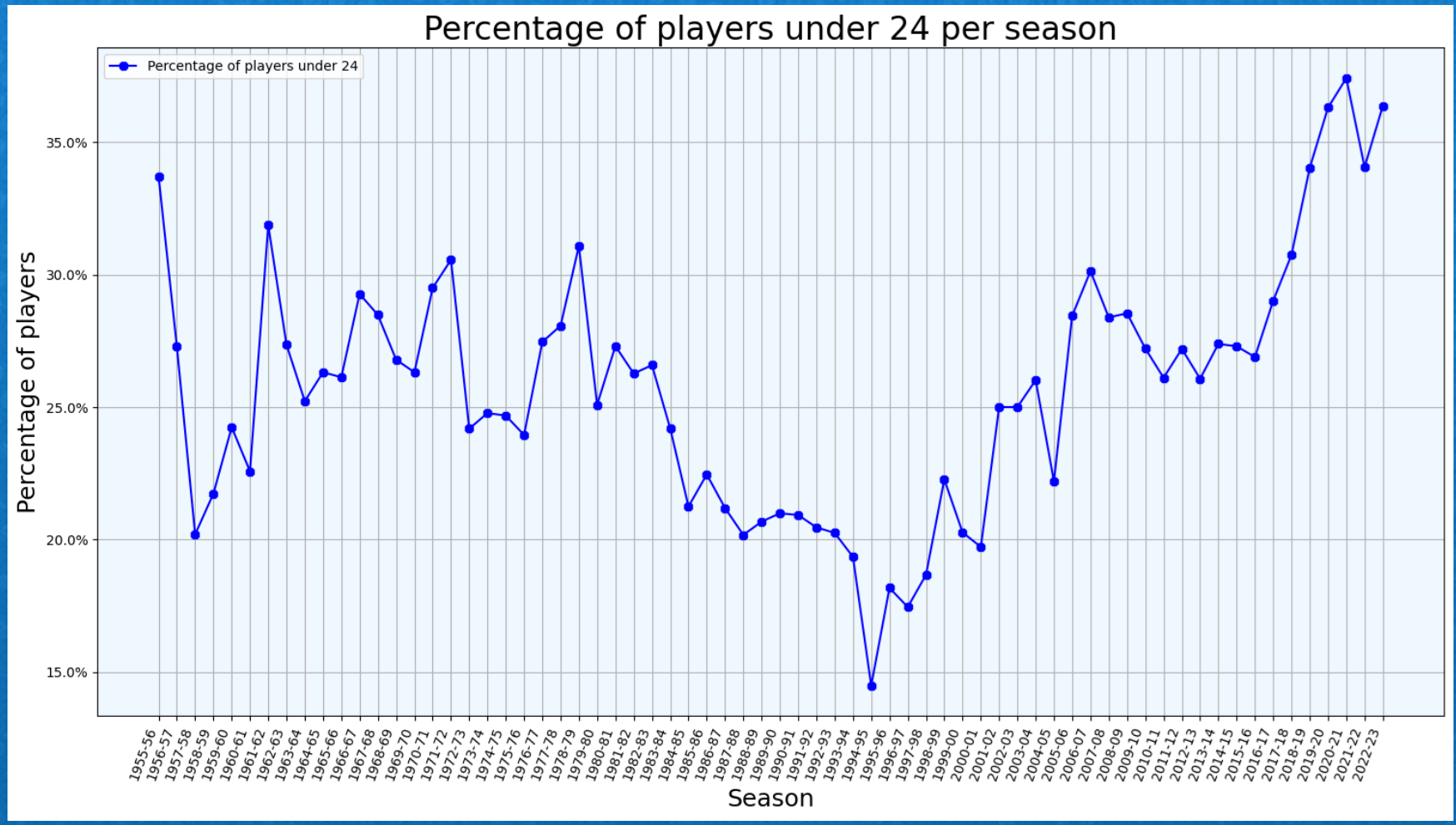
| Rk | Player | Pos | Age | Tm | G | GS | MP | FG | FGA | FG% | 3P | 3PA | 3P% | 2P | 2PA | 2P% | eFG% | FT | FTA | FT% | ORB | DRB | TRB | AST | STL | BLK | TOV | PF | PTS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Precious Achiuwa | C | 23 | TOR | 55 | 12 | 20.7 | 3.6 | 7.3 | .485 | 0.5 | 2.0 | .269 | 3.0 | 5.4 | .564 | .521 | 1.6 | 2.3 | .702 | 1.8 | 4.1 | 6.0 | 0.9 | 0.6 | 0.5 | 1.1 | 1.9 | 9.2 |
| 2 | Steven Adams | C | 29 | MEM | 42 | 42 | 27.0 | 3.7 | 6.3 | .597 | 0.0 | 0.0 | .000 | 3.7 | 6.2 | .599 | .597 | 1.1 | 3.1 | .364 | 5.1 | 6.5 | 11.5 | 2.3 | 0.9 | 1.1 | 1.9 | 2.3 | 8.6 |
| 3 | Bam Adebayo | C | 25 | MIA | 75 | 75 | 34.6 | 8.0 | 14.9 | .540 | 0.0 | 0.2 | .083 | 8.0 | 14.7 | .545 | .541 | 4.3 | 5.4 | .806 | 2.5 | 6.7 | 9.2 | 3.2 | 1.2 | 0.8 | 2.5 | 2.8 | 20.4 |
| 4 | Ochai Agbaji | SG | 22 | UTA | 59 | 22 | 20.5 | 2.8 | 6.5 | .427 | 1.4 | 3.9 | .355 | 1.4 | 2.7 | .532 | .532 | 0.9 | 1.2 | .812 | 0.7 | 1.3 | 2.1 | 1.1 | 0.3 | 0.3 | 0.7 | 1.7 | 7.9 |
| 5 | Santi Aldama | PF | 22 | MEM | 77 | 20 | 21.8 | 3.2 | 6.8 | .470 | 1.2 | 3.5 | .353 | 2.0 | 3.4 | .591 | .560 | 1.4 | 1.9 | .750 | 1.1 | 3.7 | 4.8 | 1.3 | 0.6 | 0.6 | 0.8 | 1.9 | 9.0 |
| 6 | Nickeil Alexander-Walker | SG | 24 | TOT | 59 | 3 | 15.0 | 2.2 | 5.0 | .444 | 1.0 | 2.7 | .384 | 1.2 | 2.3 | .515 | .547 | 0.7 | 1.0 | .667 | 0.3 | 1.5 | 1.7 | 1.8 | 0.5 | 0.4 | 0.9 | 1.5 | 6.2 |
| 6 | Nickeil Alexander-Walker | SG | 24 | UTA | 36 | 3 | 14.7 | 2.3 | 4.7 | .488 | 1.0 | 2.4 | .402 | 1.3 | 2.3 | .578 | .591 | 0.8 | 1.1 | .692 | 0.2 | 1.4 | 1.6 | 2.1 | 0.7 | 0.4 | 1.3 | 1.6 | 6.3 |
| 6 | Nickeil Alexander-Walker | SG | 24 | MIN | 23 | 0 | 15.5 | 2.1 | 5.4 | .384 | 1.1 | 3.1 | .361 | 1.0 | 2.3 | .415 | .488 | 0.6 | 0.9 | .619 | 0.3 | 1.5 | 1.8 | 1.4 | 0.3 | 0.3 | 0.4 | 1.3 | 5.9 |
| 7 | Grayson Allen | SG | 27 | MIL | 72 | 70 | 27.4 | 3.4 | 7.7 | .440 | 2.0 | 5.1 | .399 | 1.4 | 2.7 | .518 | .571 | 1.6 | 1.8 | .905 | 0.8 | 2.4 | 3.3 | 2.3 | 0.9 | 0.2 | 1.0 | 1.6 | 10.4 |

- Each row is the per-game stats of a player in one season for one team (TOT = all teams combined, for traded players)

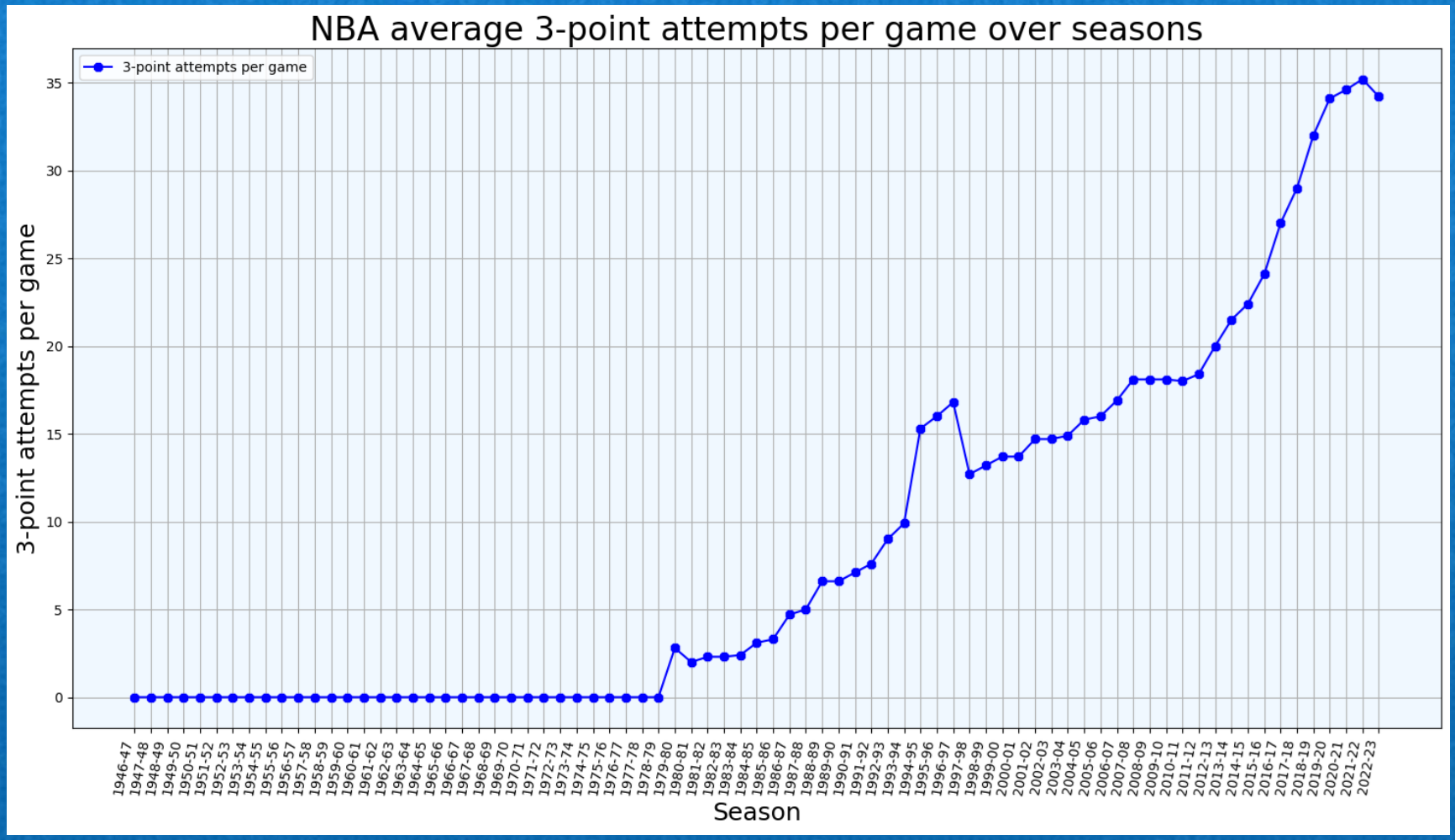
# VISUALIZATION OVER TIME



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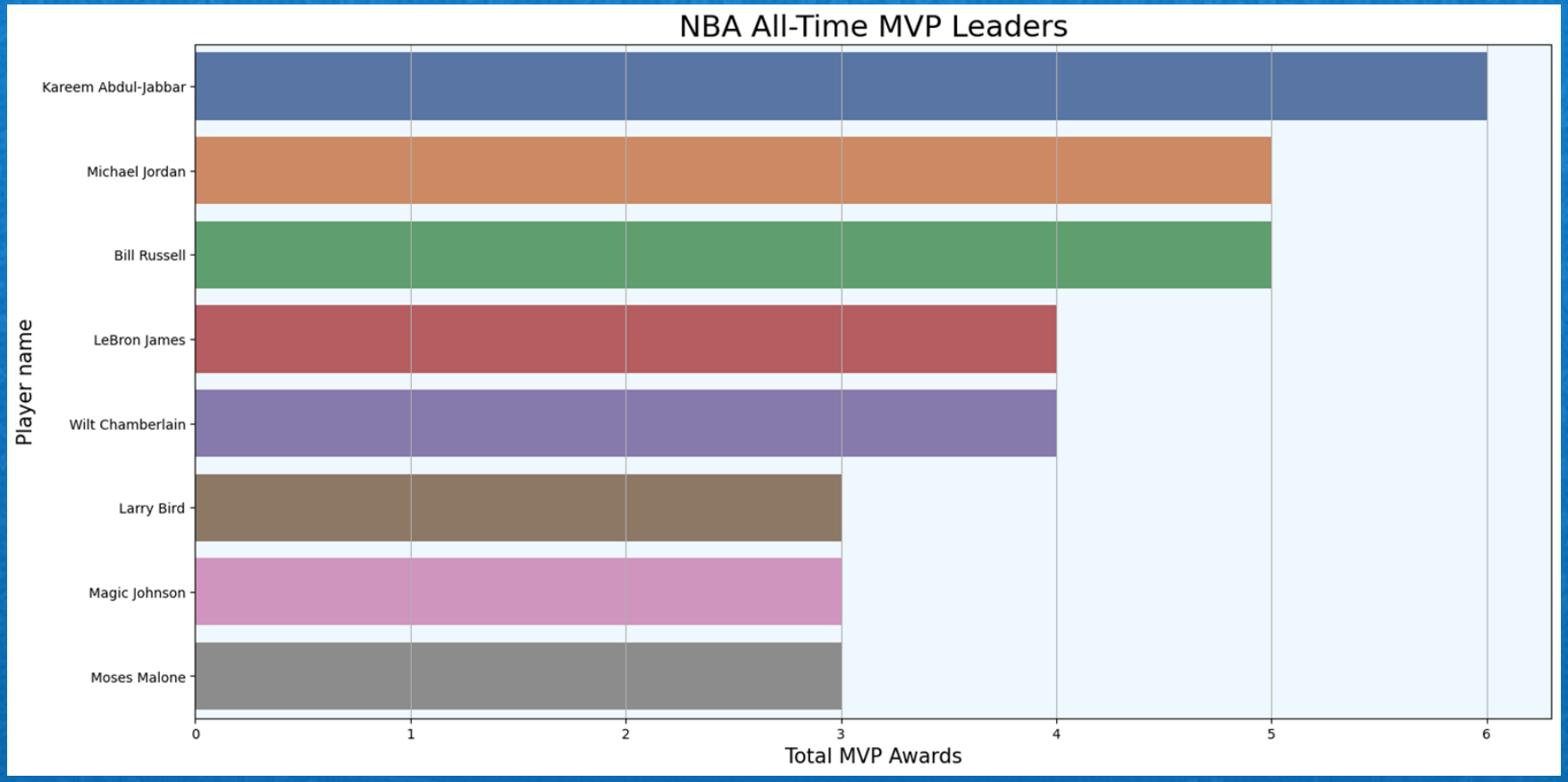
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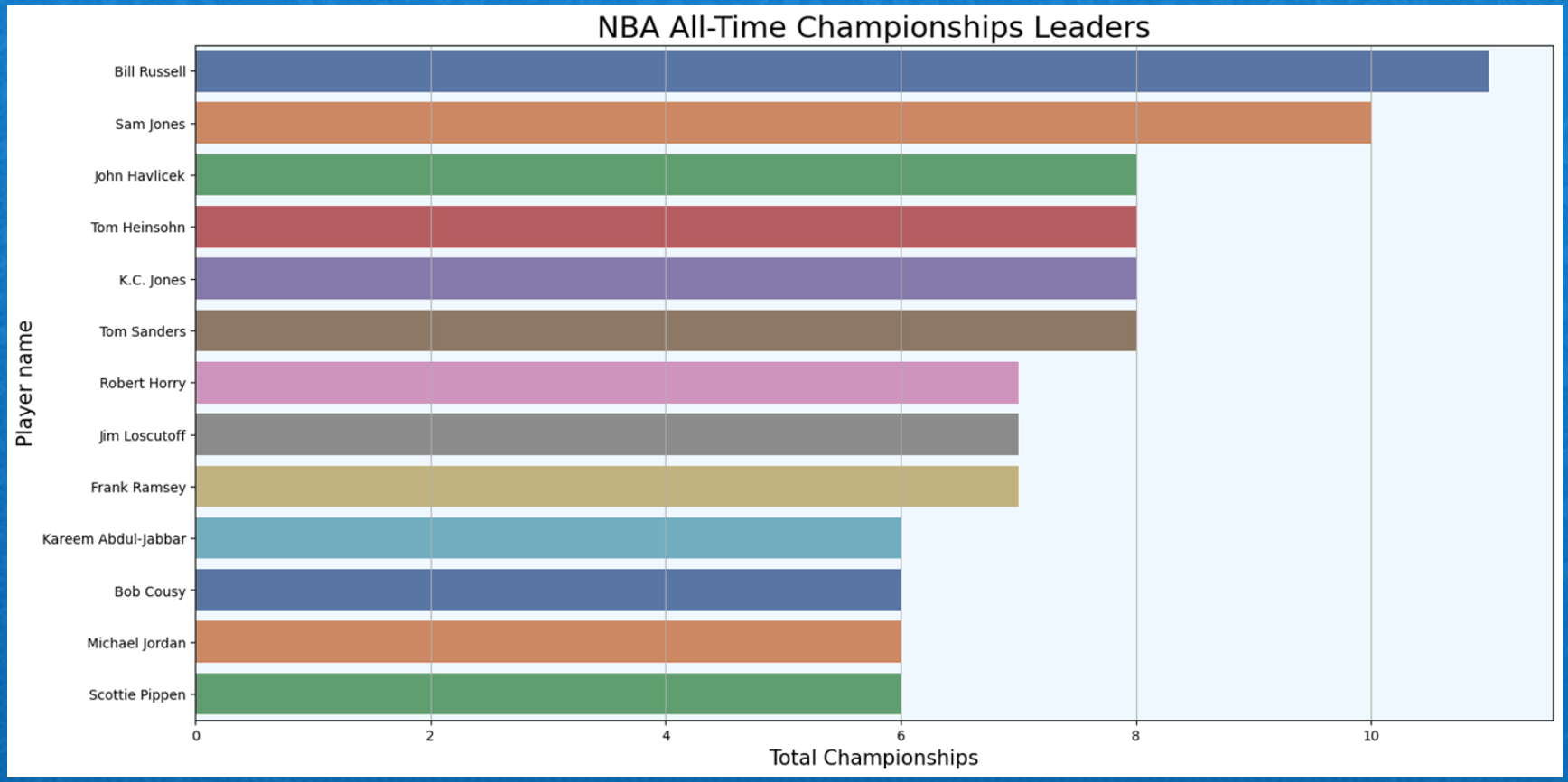
# CONCLUSIONS

- After the 1990s with the traditional big men and post-up game focus, the NBA is becoming a young players' league
- The game is evolving, becoming quicker, more possessions (early shot-clock offence) and more 3-point shots (even in transition)
- As a result, young players can make a big difference with their athletic ability and stamina
- The 3-point shot has become a vital skill for players' and teams' success

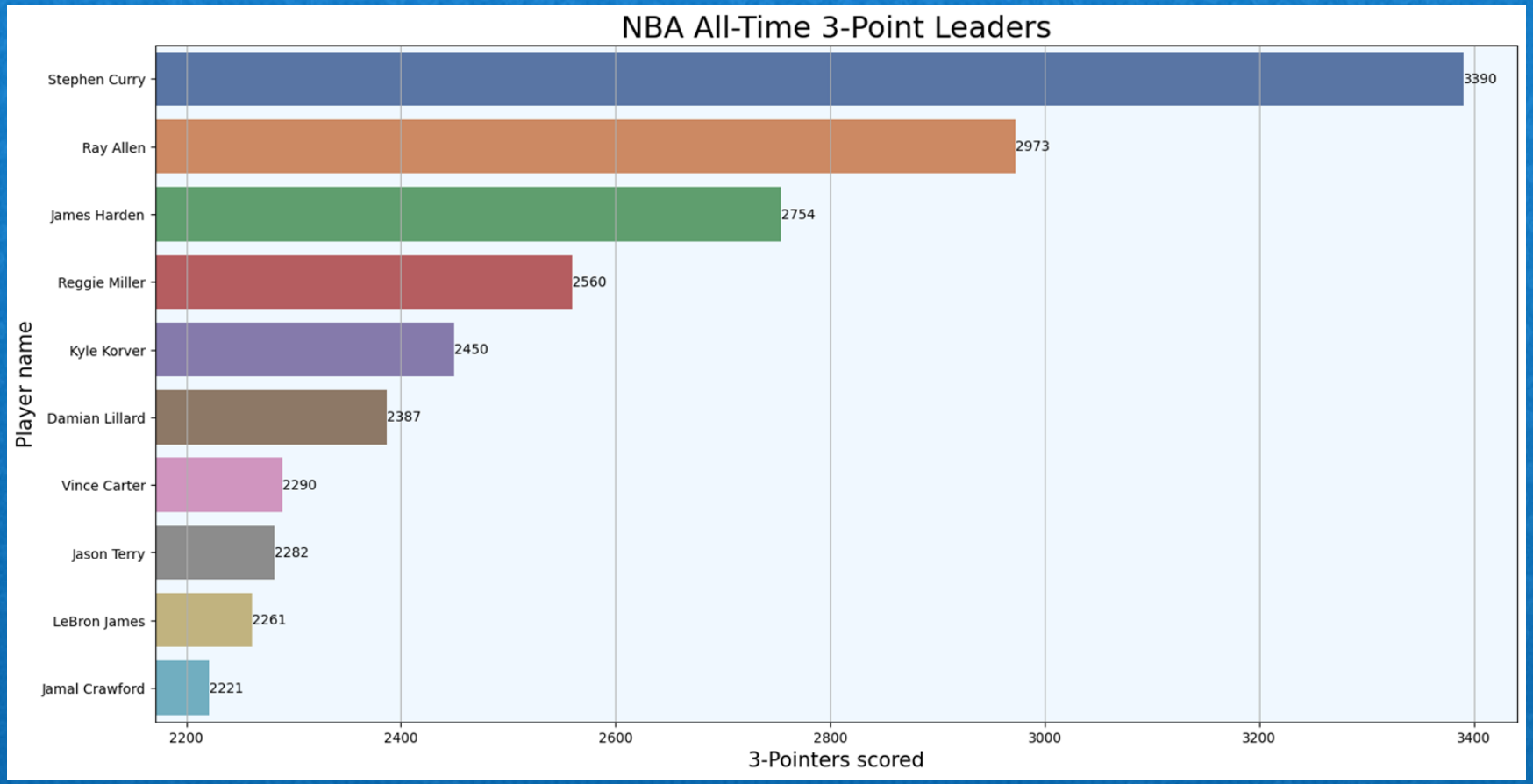
# Career Stats Visualization



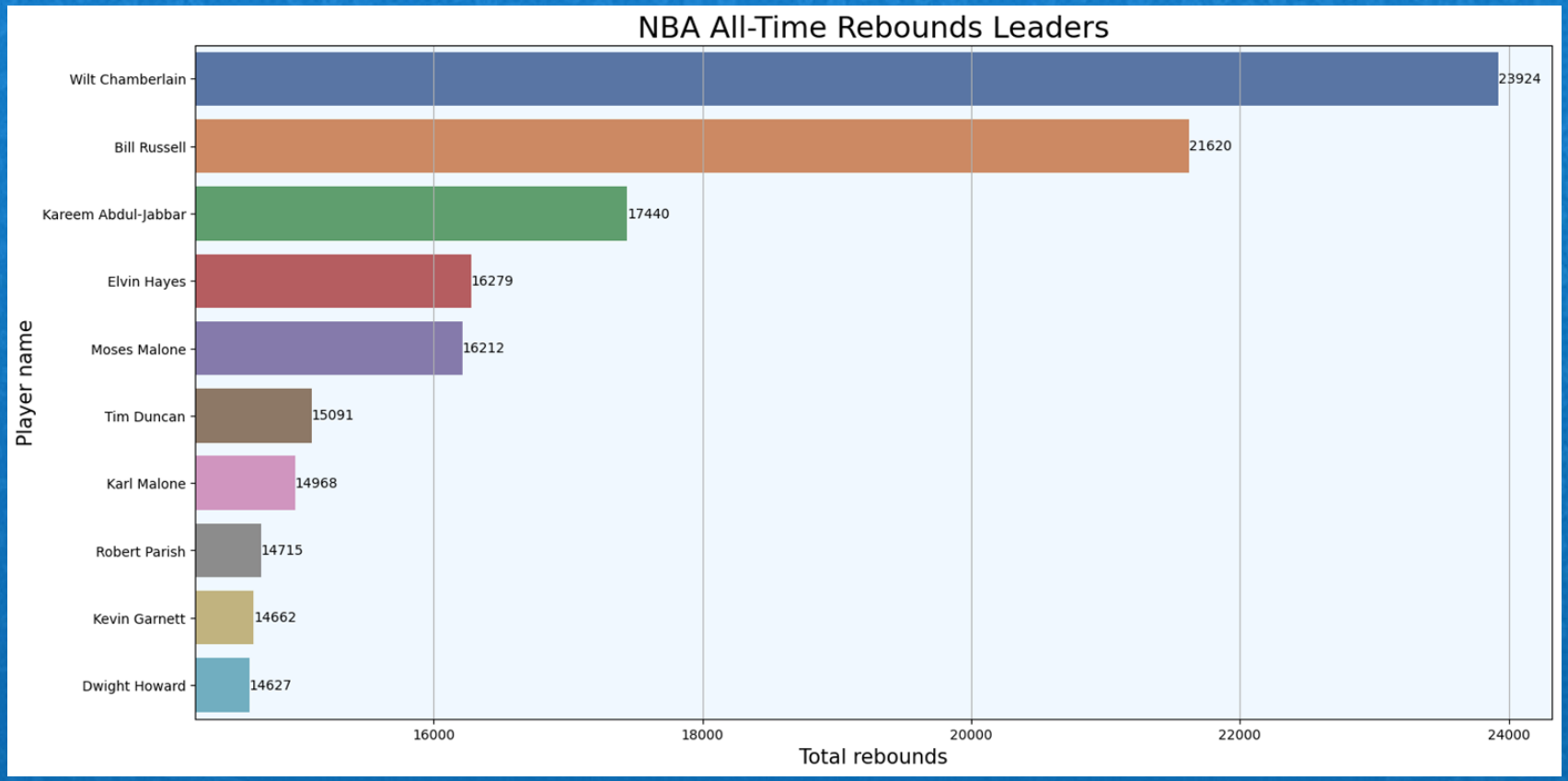
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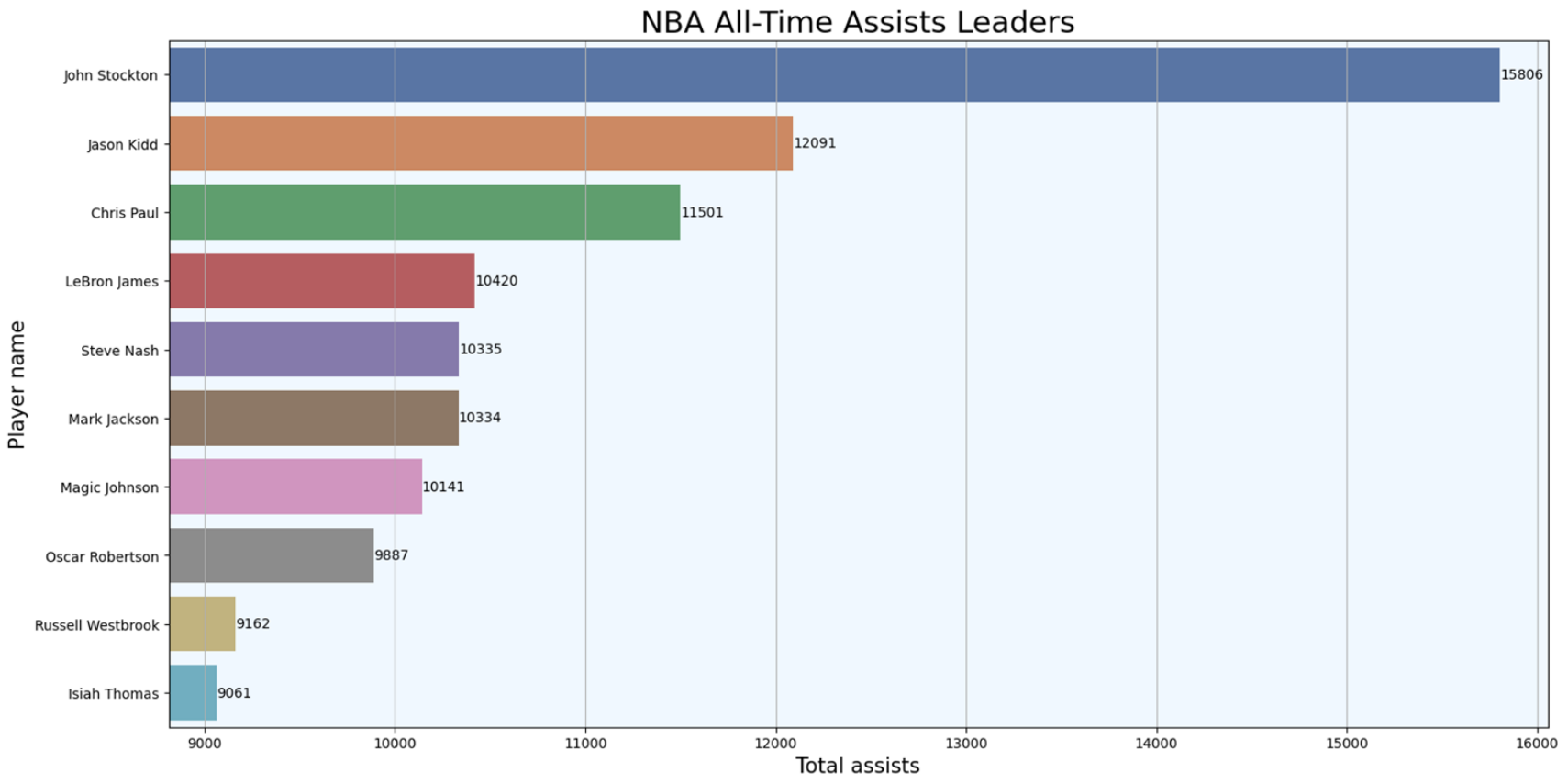
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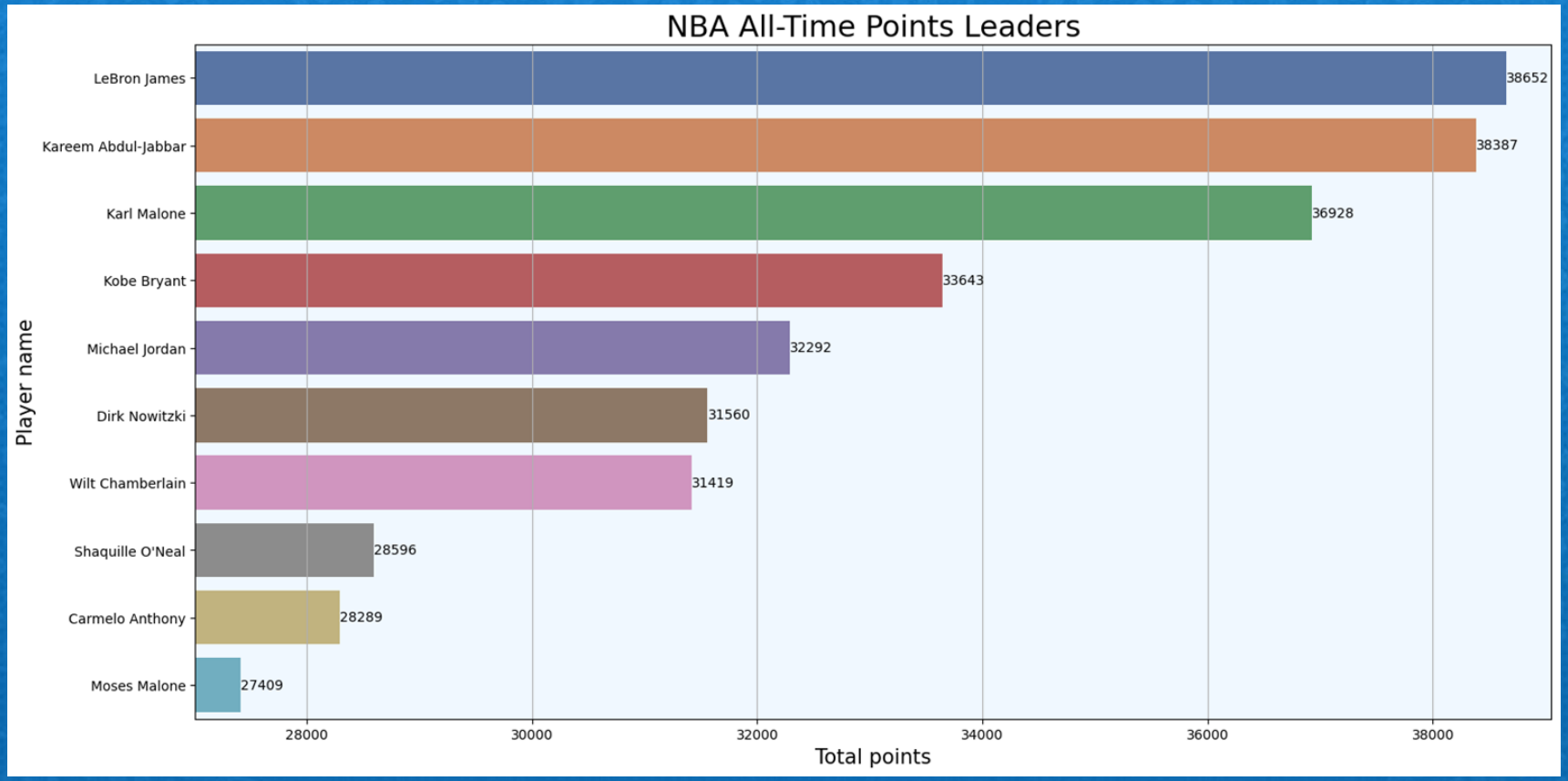
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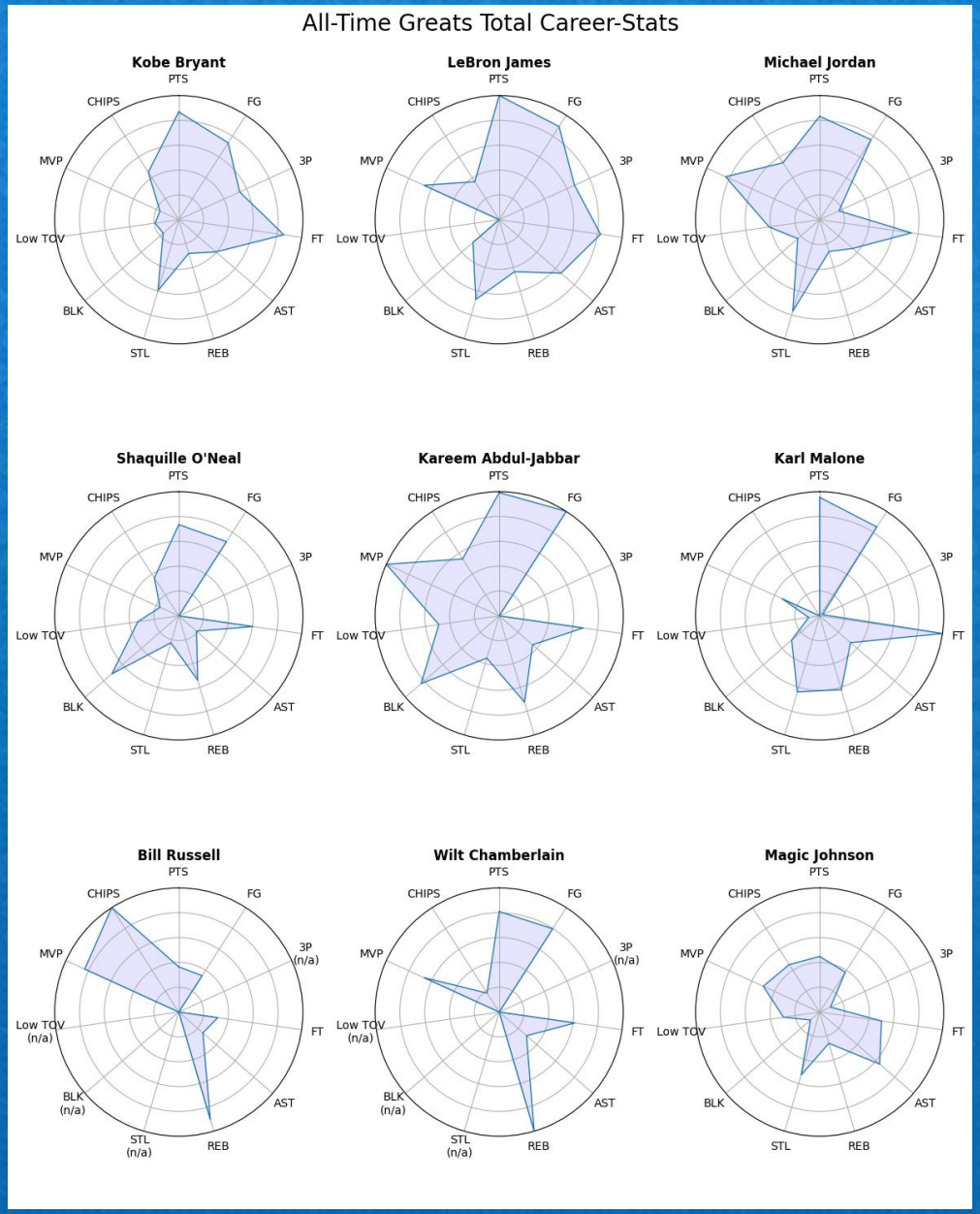
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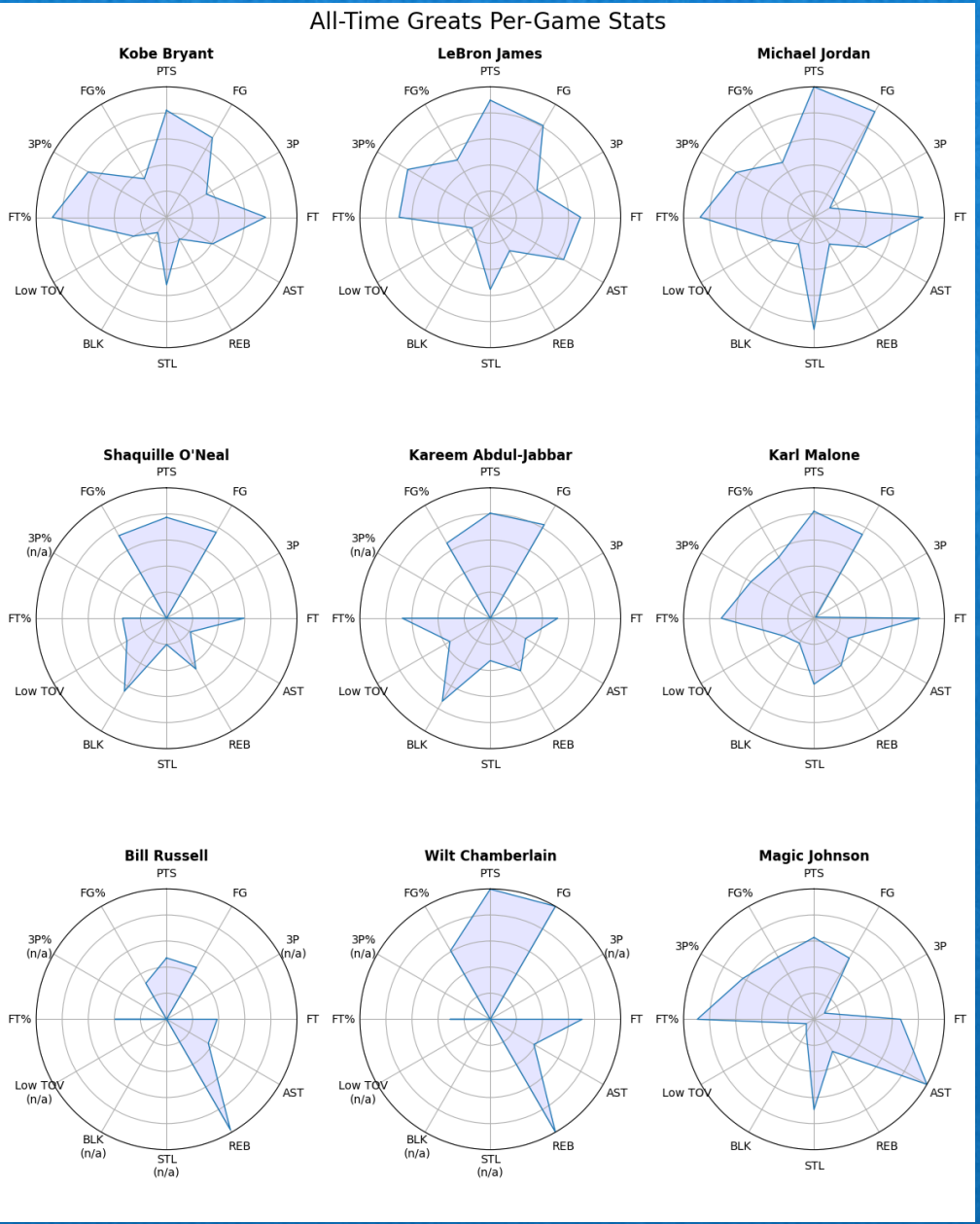
# Career Stats Visualization



# GOAT Comparison



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# CONCLUSIONS

- The total-career stats radar-plots are great to see where each great dominated during his career.
- Kareem Abdul-Jabbar and LeBron James have had the greatest NBA careers stat-wise.
- The per-game radar plots show how the greats affected their games, with Michael Jordan the most complete.
- Michael Jordan and LeBron James are head-to-head the greatest players of all time.

# REFERENCES

- Complete Jupyter Notebook Analysis
- https://www.basketball-reference.com/

